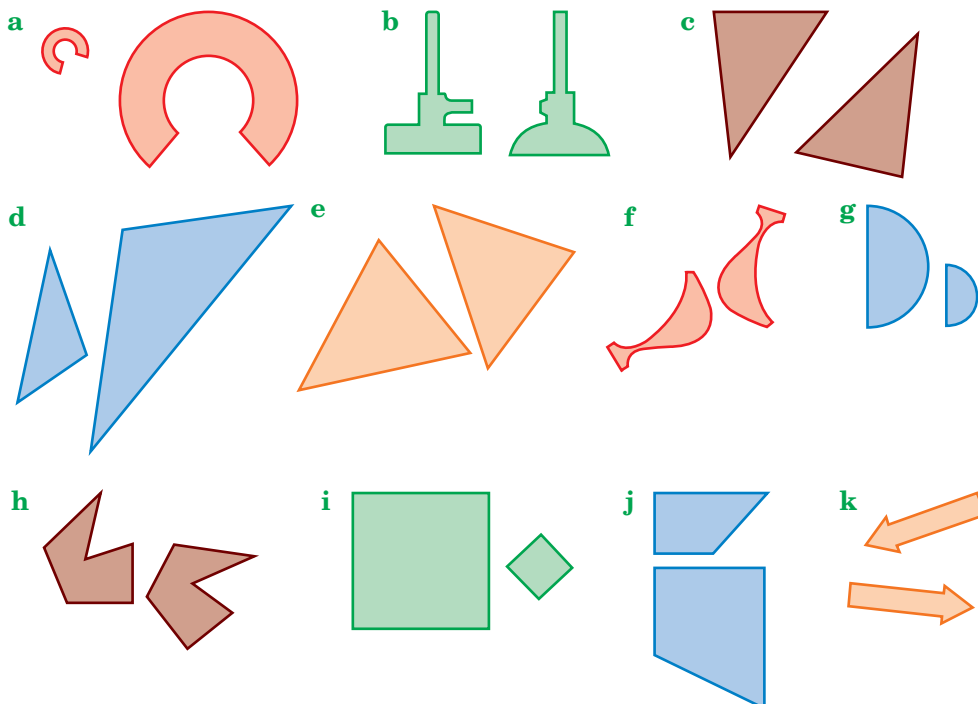


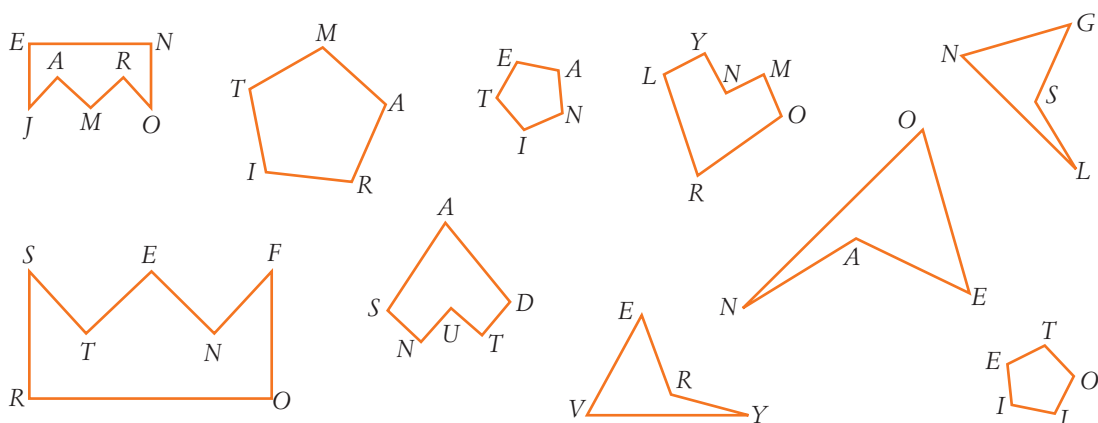
Extra questions (Year 10+ 10A)

Exercise 8.2 Similarity and congruence

1 State whether each pair of shapes is congruent, similar or neither.



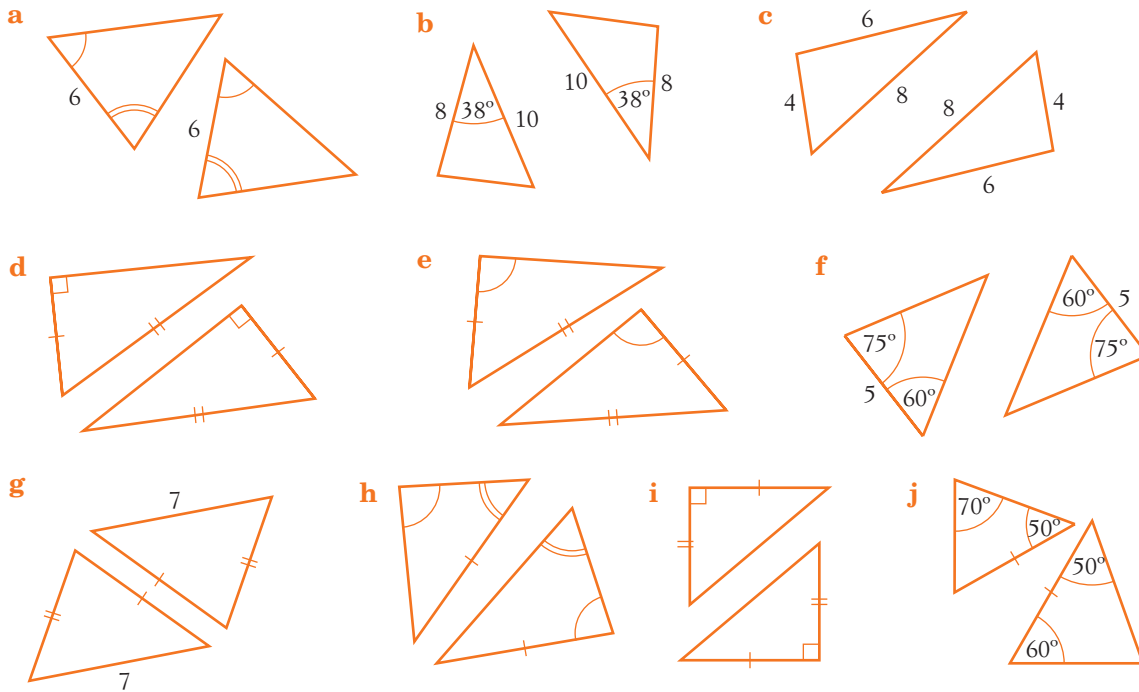
2 Name the pairs of similar and congruent shapes below. Work out the scale factor for similar shapes.



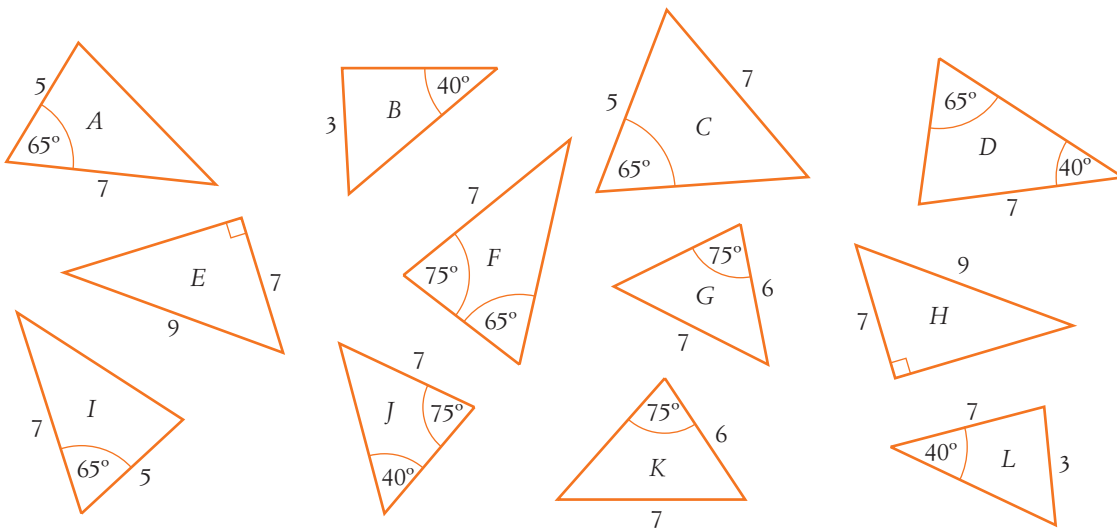
3 Calculate the scale factor for each pair of similar shapes you found in question 1.

4 Calculate the scale factor for each pair of similar shapes you found in question 2.

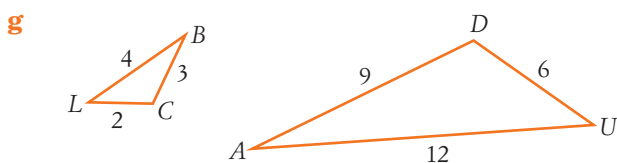
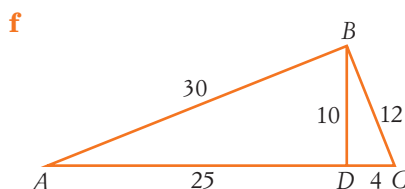
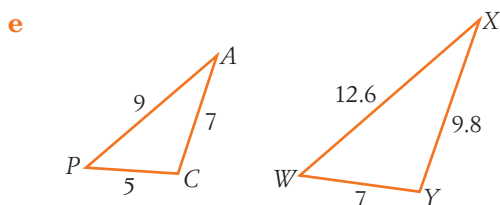
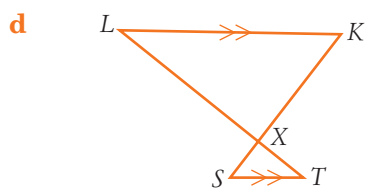
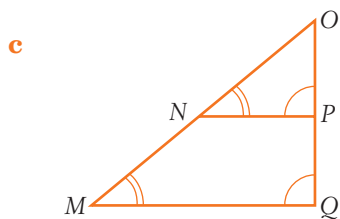
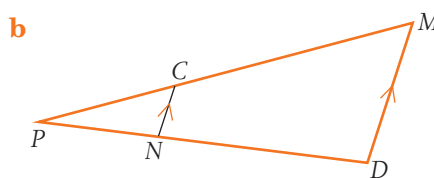
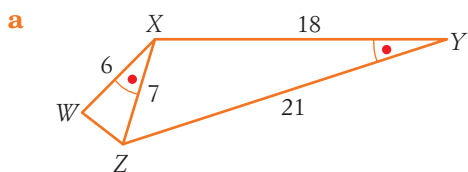
5 State whether each of these pairs of triangles is congruent and state the test used.



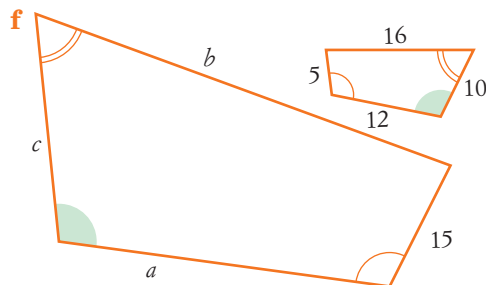
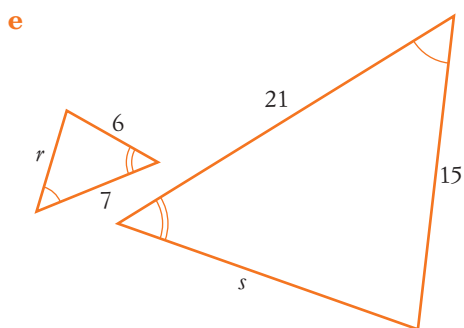
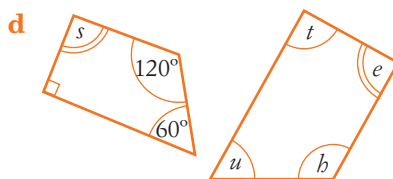
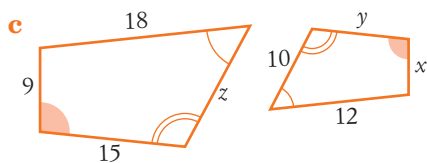
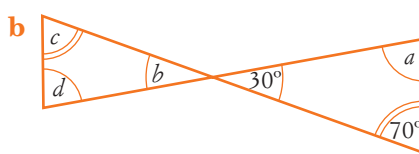
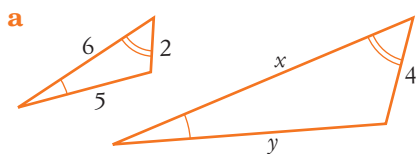
6 Choose the pairs of congruent triangles below and state the test used in each case.



7 State whether each pair of triangles below is similar, giving reasons.

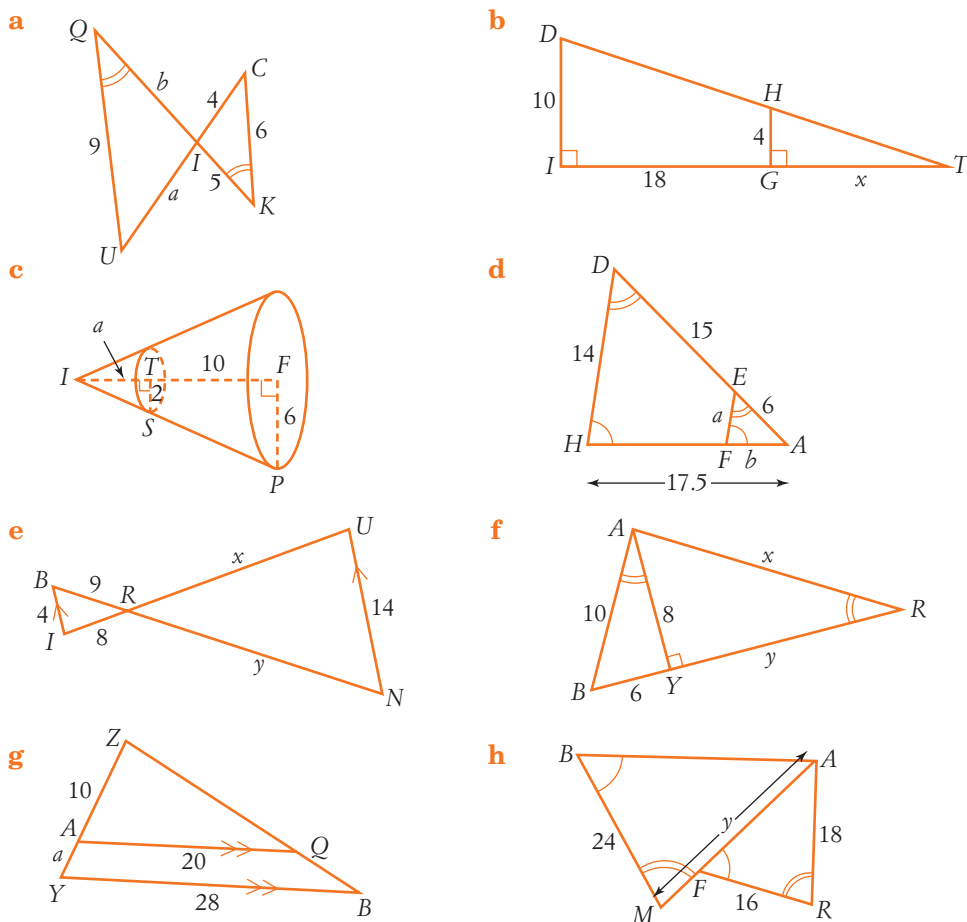


8 The pairs of figures below are similar. Calculate the values of the pronumerals.

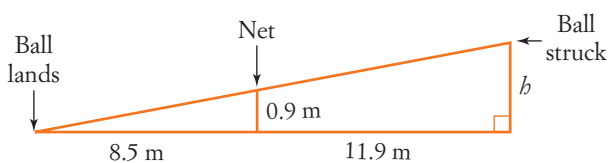


- 9 **a** Draw rectangle $PQRS$ with $PQ = 40$ mm and $QR = 70$ mm.
b Draw rectangle $KLNM$ with $KL = 126$ mm and $LM = 72$ mm.
c Are the two rectangles similar? Why?
d Complete: $PQRS \parallel \dots$

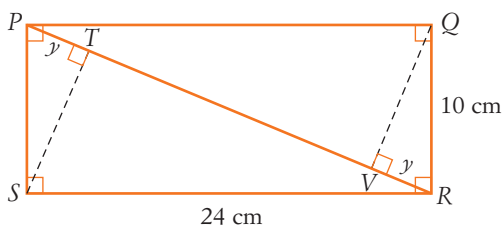
10 Find the unknowns in each of the following, giving reasons.



11 A tennis ball is served and it just clears the net. It lands 8.5 m inside the opponent's court. Use the diagram and your knowledge of similar triangles to find the height at which the ball was struck. (Assume that the ball travels in a straight line.)



12 PQRS is a rectangle with dimensions as shown.



- Find the length of PR .
- Show that $\triangle PTS$ and $\triangle PSR$ are similar.
- Find y and hence find the length of TV .

Answers

- 1 Congruent: **c, f, h, k**
Similar (but not congruent), with scale factors: **a, d, g, i**
Neither (different): **b, e, j**
- 2 Congruent: $TEANI \equiv TOLIE$, $LYNMOR \equiv DTUNSA$, $NGSL \equiv VERY$
Similar, with scale factors: $NGSL$ and $VERY \parallel OEAN$; $TMARI \parallel TEANI$ and $TOLIE$; $ENORMAJ \equiv ORSTENF$
- 3 Scale factors: $\mathbf{a} \times 4$, $\mathbf{d} \times 2$, $\mathbf{g} \times \frac{1}{2}$, $\mathbf{i} \times \frac{1}{3}$
- 4 Scale factors: $NGSL$, $VERY$ and $OEAN$, $\times 1.5$; $TMARI$, $TEANI$ and $TOLIE$, $\times \frac{1}{2}$; $ENORMAJ \equiv ORSTENF$, $\times 2$
- 5 Congruent:
a ASA **b** SAS **c** SSS **d** RHS
f ASA, **g** SSS, **i** SAS, **j** ASA
Not necessarily congruent: **e, h**
- 6 $A \equiv I$, SAS; $D \equiv F$, AAS; $E \equiv H$, RHS. Others are not necessarily congruent.
- 7 **a** $\triangle ZWX \parallel \triangle ZXY$ (SAS) **b** $\triangle PNC \parallel \triangle PDM$ (AAA)
c $\triangle ONP \parallel \triangle OMQ$ (AAA) **d** $\triangle LKX \parallel \triangle TSX$ (AAA)
e $\triangle APC \parallel \triangle XWY$ (SSS) **f** $\triangle ABD \parallel \triangle BCD$ (SSS)
g $\triangle BCL \parallel \triangle ADU$ (SSS)
- 8 **a** $x = 12, y = 10$ **b** $b = 30^\circ, c = 70^\circ, d = 80^\circ$
c $x = 6, y = 10, z = 15$ **d** $e = s = 90^\circ, h = 120^\circ, t = 90^\circ, u = 60^\circ$
e $r = 5, s = 18$ **f** $a = 36, b = 48, c = 30$
- 9 **a** Teacher to check.
b Teacher to check.
c The two rectangles are similar because the ratio of corresponding sides is equal.
d $PQRS \parallel NKLM$
- 10 **a** $\triangle QUI \parallel \triangle KCI$ (AAA), $a = 6, b = 7.5$ **b** $\triangle DTI \parallel \triangle HTD$ (AAA), $x = 12$
c $\triangle ITS \parallel \triangle IFP$ (AAA), $a = 5$ **d** $\triangle DAH \parallel \triangle EAF$ (AAA), $a = 4, b = 5$
e $\triangle BRI \parallel \triangle NRU$ (AAA), $x = 28, y = 31.5$ **f** $\triangle AYB \parallel \triangle RYA$ (AAA), $x = 13\frac{1}{3}, y = 10\frac{2}{3}$
g $\triangle ZAQ \parallel \triangle ZYB$ (AAA), $a = 4$ **h** $\triangle BAM \parallel \triangle FAR$ (AAA), $y = 27$
- 11 2.16 m
- 12 **a** 26 cm
b $\angle PTS = \angle PSR = 90^\circ$
 $\angle SPT = \angle RPS$ (common angle)
so $\angle PST = \angle PRS$ (angle sum of triangle )
so $\triangle PTS \parallel \triangle PSR$ (AAA)
c $y \approx 3.8$ cm, $TV \approx 18.3$ cm